Regulatory Separation and Pharmaceutical Innovation: Evidence from China's MAH Reform

Abstract

This paper develops a partial-equilibrium mechanism model of how China's Marketing Authorization Holder (MAH) system changes the value of advancing viable drug projects and their commercialization organization. Developers choose one common original-drug innovation investment / project-advancement intensity before project characteristics and organizational routes are realized. MAH makes retained entrusted manufacturing legally available through one institutional wedge, while route choice remains deterministic and the developer retains authorization-holder responsibility. Internal and entrusted production are derived from distinct technologies, and the price of qualified manufacturing services is endogenous. Reform effects are confined to project–developer pairs for which the new route improves on the best old option and can be attenuated by service scarcity. The baseline does not treat MAH as a demand, scientific-productivity, downstream-success, or patenting shock, and it imposes no original-versus-incremental response ranking. A separate resource-allocation extension endogenizes upstream research and patent applications. It predicts that a strict commercialization-value gain raises development but can reduce research and patenting when a common innovation-resource constraint binds; without that constraint, research does not fall.

Keywords: MAH; pharmaceutical innovation; commercialization frictions; entrusted production; resource allocation; patenting.
JEL codes: D22; L65; O31; O38; I18.

1 Introduction

China's MAH system separates the right to hold marketing authorization from a strict requirement of in-house production. This institutional change is potentially important for drug development because many research-originating firms can generate valuable projects but lack complete manufacturing capacity. The central mechanism studied here is therefore not demand expansion or scientific productivity. It is an expected-commercialization-value channel: when a firm anticipates that an advancing project can later use qualified external manufacturing while the firm retains authorization, the expected value of advancing that project may increase.

The mechanism builds on two established ideas. First, pharmaceutical innovation responds to expected downstream rewards (Schmookler, 1966; Acemoglu and Linn, 2004; Finkelstein, 2004; Dubois et al., 2015; Budish et al., 2015). Second, an innovator's ability to appropriate returns depends on complementary assets and commercialization strategy (Teece, 1986; Arora et al., 2001; Gans and Stern, 2003). MAH fits this second logic because it changes how an innovator may access manufacturing capability while retaining the authorization asset.

Recent evidence on China's pharmaceutical reforms helps separate this route mechanism from two nearby channels. Jia et al. (2023) study the 2015 drug-review reform and emphasize approval delay, review capacity, and the composition of firms and products entering the pipeline. Barwick et al. (2026) study the National Reimbursement Drug List and emphasize demand and market-size incentives. Those channels can coexist with MAH, but they are not the MAH mechanism in this paper. The baseline holds scientific productivity and the market-return environment fixed and asks whether a regulated holder–producer separation route changes expected commercialization value.

Closely related evidence also shows why project advancement and upstream patenting must remain distinct. Gu (2024) finds that allowing production outsourcing raises clinical-trial registrations among innovators without production facilities, while patent applications fall among firms with fewer financial resources. The paper also reports heterogeneous responses across development stages and drug classes. These findings motivate empirical heterogeneity tests and the separate resource-allocation extension developed below. The extension makes patent applications endogenous without turning them into a baseline outcome or a direct MAH shock. It also preserves the baseline's refusal to impose an original-versus-incremental ranking.

For a development-economics audience, the relevant bottleneck is a regulated complementary asset. MAH can relax a manufacturing bottleneck for research-oriented firms while preserving holder responsibility. Scarcity in qualified production support can still attenuate the effect.

The model is deliberately narrow. It is not a full dynamic structural model of the pharmaceutical industry. It does not solve product-market clearing, an input market for project advancement, free entry, exit, or an invariant firm distribution. The baseline closes only the market directly tied to the MAH mechanism: qualified manufacturing services. Reform-induced demand for retained entrusted production can raise the equilibrium service price p_m^*, so the model does not mechanically imply a positive effect for every project or developer.

The contribution is to organize the mechanism in a way that is internally consistent with feasible data. The theory separates project advancement from planning-stage project arrival, deterministic organizational assignment, observed holder–producer separation, and realized products. The empirical role of the model is correspondingly modest: it specifies empirically interpretable interfaces for project-level organization, retained outcomes, and service scarcity without claiming that approval-side data separately identify every primitive friction.

The model is presented in three layers. The main text gives the shortest complete commercialization baseline, then adds a separately tagged research–development allocation extension. The technical appendix records primitives, payoff accounting, pricing derivations, deterministic sorting, project advancement, qualified-capacity equilibrium, comparative statics, observed outcomes, and explicitly separated extensions.

2 Institutional Background

The relevant institutional point is timing. Under China's drug-registration framework, a firm need not already be the final commercial holder when it begins to advance a viable project. The economic choice is forward-looking: a firm invests today while anticipating the manufacturing and commercialization environment it will face if a project advances. Production support and manufacturing readiness must be planned before final approval is observed. After obtaining a Drug Approval License, the applicant becomes the MAH (National Medical Products Administration, 2022a). The model therefore treats MAH as changing the anticipated availability and value of a future organizational route, not current scientific productivity or downstream success.

The MAH reform creates a regulated holder-producer separation route. The 2016 pilot allowed eligible applicants in selected regions to become holders under the pilot arrangement (General Office of the State Council of the People's Republic of China, 2016; China Food and Drug Administration, 2016). The later legal framework consolidated the principle that an MAH may manufacture by itself or entrust production to a qualified manufacturer (National Medical Products Administration, 2019). Subsequent NMPA rules require holder-side quality responsibility, assessment of the entrusted manufacturer, quality and entrustment agreements, and ongoing supervision (National Medical Products Administration, 2022c,b, 2023).

These rules imply two modeling restrictions. First, entrusted production is not anonymous outsourcing or transfer. It is a regulated retained route in which authorization remains with the developer while production is performed by a qualified external manufacturer. Second, MAH does not eliminate holder responsibility. The entrusted payoff therefore retains a holder-side burden μ_E, while the institutional wedge $\tau_E(M)$ governs effective route availability. The downstream realization probability $s(q)$ is route independent and invariant to MAH in the baseline.

3 Model

3.1 Environment and timing

There is a continuum of developers. Developer i has predetermined research capability $a_i > 0$ and internal manufacturing capability $k_i > 0$, distributed according to $H(a, k)$. A viable project that reaches commercialization-relevant planning draws value $q > 0$ and manufacturing requirement $m > 0$ from the exogenous distribution $F(q, m)$. For empirical decompositions only,

$$F(q,m) = \sum_{g \in \{O, \text{Inc}\}} \rho_g F_g(q,m), \qquad \rho_g \geq 0, \quad \sum_g \rho_g = 1.$$

The class label g creates no class-specific control.

The institutional regime is $M \in \{0, 1\}$. The pre-MAH regime sets the barrier to retained entrusted manufacturing at infinity; the MAH regime makes it finite:

$$\tau_E(0) = +\infty, \qquad \tau_E(1) = \bar{\tau}_E < +\infty, \qquad \bar{\tau}_E \geq 0. \tag{1}$$

The available routes are internal manufacturing I, retained entrusted manufacturing E, non-retained transfer T, and abandonment or delay A. Under E, the developer remains the authorization holder and a qualified external producer manufactures. Thus E is not transfer.

Before project characteristics and routes are realized, developer i chooses one common $x_i \geq 0$, interpreted as original-drug innovation investment / project-advancement intensity. It is broader than pure clinical development, but excludes basic research, compound discovery, patent-generating effort, and patent applications. The expected measure of viable projects reaching route planning is

$$\lambda_i^{\text{plan}} = a_i x_i. \tag{2}$$

Timing is as follows. First, M is observed and developers anticipate the market-consistent qualified-service price. Second, they choose x_i. Third, planning-stage projects draw (q, m) and choose a route. Fourth, downstream realization occurs with exogenous probability $s(q)$. Finally, aggregate capacity demand and supply must validate the anticipated service price. The last step is an equilibrium consistency condition, not a late unanticipated shock. Upstream research and patent generation may help determine the predetermined a_i, but they are outside the baseline decision problem.

3.2 Commercialization technologies

Conditional on successful commercialization, residual demand is $y(p;q) = Aqp^{-\varepsilon}$, where $A > 0$ and $\varepsilon > 1$. Product-price optimization gives

$$p^*(c) = \frac{\varepsilon}{\varepsilon - 1} c$$

and the gross operating present value

$$R(q,c) = \frac{Aq}{1-\beta\varphi} \frac{(\varepsilon-1)^{\varepsilon-1}}{\varepsilon^{\varepsilon}} c^{1-\varepsilon}, \qquad \beta \in (0,1), \quad \varphi \in [0,1). \tag{3}$$

The appendix derives the FOC, SOC, and accounting. In particular, $R_q > 0$ and $R_c < 0$.

Internal production uses marginal cost $c_I(m, k_i)$, with $c_{I,m} > 0$ and $c_{I,k} < 0$, and setup cost $F_I(m, k_i)$, with $F_{I,m} > 0$ and $F_{I,k} < 0$ on the feasible domain. Internal production is unavailable when $k_i < \underline{k}(m)$, represented by $F_I(m, k_i) = +\infty$. Its route value is

$$W_i^I(q,m) = s(q)R(q, c_I(m, k_i)) - F_I(m, k_i). \tag{4}$$

Entrusted production uses external marginal cost $c_E(m)$, requires $b(m) > 0$ units of qualified capacity with $b'(m) > 0$, incurs setup cost $F_E(m)$, and leaves the holder with burden $\mu_E \geq 0$. At the qualified-capacity price p_m,

$$\begin{aligned} W_i^E(q,m;M,p_m) = s(q)R(q, c_E(m)) - F_E(m) - p_m b(m) \\ - \mu_E - \tau_E(M). \end{aligned} \tag{5}$$

Each monetary cost enters once. The transfer and abandonment values are

$$W^T(q,m) = T(q,m), \qquad W^A = 0. \tag{6}$$

3.3 Organization and project advancement

At a fixed p_m, the project chooses deterministically:

$$\begin{aligned} W_i(q,m;M,p_m) &= \max\{W_i^I, W_i^E, W^T, W^A\}, \\ r_i^*(q,m;M,p_m) &\in \arg\max_{r\in\{I,E,T,A\}} W_i^r. \end{aligned} \tag{7}$$

Continuous heterogeneity makes ties a zero-measure event; no logit or inclusive value is used.

The internal-minus-entrusted gap is

$$\begin{aligned}\Delta_{IE}(k_i) = s(q)[R(q, c_I(m, k_i)) - R(q, c_E(m))] - F_I(m, k_i) \\ + F_E(m) + p_m b(m) + \mu_E + \tau_E(M).\end{aligned} \tag{8}$$

Holding (q, m, M, p_m) fixed on the internally feasible domain,

$$\Delta_{IE,k} = s(q) R_c(q, c_I)\, c_{I,k} - F_{I,k} > 0.$$

When the endpoint crossing conditions hold, a unique conditional cutoff $k^*(q, m; p_m, M)$ satisfies $\Delta_{IE}(k^*) = 0$. Conditional on I and E dominating T and A, developers below the cutoff select E, while those above it select I. Transfer and abandonment remain valid outside options.

Expected optimized value per planning-stage project is

$$\Omega_i(M, p_m) = \int W_i(q, m; M, p_m)\, dF(q, m). \tag{9}$$

Developer i solves

$$\max_{x_i \geq 0} \left\{ \beta a_i x_i \Omega_i(M, p_m) - \frac{\kappa}{1+\nu} x_i^{1+\nu} \right\}, \qquad \kappa > 0, \quad \nu > 0. \tag{10}$$

Strict concavity gives the unique solution

$$x_i^*(M, p_m) = \left[\frac{\beta a_i}{\kappa} \Omega_i(M, p_m) \right]^{1/\nu}, \tag{11}$$

including $x_i^* = 0$ when $\Omega_i = 0$. The binary reform affects x_i^* only through Ω_i. At a fixed p_m, its effect is strict only if E improves on the pre-MAH maximum on a positive-probability project set.

3.4 Qualified manufacturing-service equilibrium

Qualified supplier j, with efficiency z_j, chooses capacity $s_j \geq 0$:

$$\max_{s_j \geq 0} \{ p_m s_j - \Psi(s_j; z_j) \}. \tag{12}$$

With $\Psi_{ss} > 0$, positive-price capacity is uniquely determined by $p_m = \Psi_s(s_j^*; z_j)$, and aggregate supply is

$$S_m(p_m) = \int s_j^*(p_m, z)\, dH_C(z).$$

Expected entrusted capacity per planning-stage project is

$$\chi_i^E(p_m; M) = \int b(m)\mathbf{1}\{r_i^*(q, m; M, p_m) = E\}\, dF(q, m). \tag{13}$$

Study-related and total demand are

$$\begin{aligned} D_m^{\text{MAH}}(p_m; M) &= \int a_i x_i^*(M, p_m)\chi_i^E(p_m; M)\, dH(a, k), \\ D_m(p_m; M) &= D_m^B(p_m) + D_m^{\text{MAH}}(p_m; M). \end{aligned} \tag{14}$$

The demand expression includes both feedbacks: price changes expected value and hence advancement, and it changes deterministic route selection. Continuous heterogeneity makes aggregate demand continuous even though individual indicators can jump.

The qualified-capacity market clears at the unique price

$$D_m(p_m^*; M) = S_m(p_m^*). \tag{15}$$

Supply is strictly increasing and demand weakly decreasing. Positive background demand gives excess demand at zero, while entrusted and background demand vanish and supply diverges at high prices. These conditions give existence and uniqueness. If post-MAH study demand is positive at the old price, $p_m^*(1) > p_m^*(0)$: CMO scarcity attenuates the reform rather than amplifying it through a direct price reduction.

The baseline partial equilibrium contains exactly

$$\left\{p_m^*,\ x_i^*,\ r_i^*(q, m),\ s_j^*\right\}_{i,j}. \tag{16}$$

No entry, labor, capital, product-market aggregate, or welfare closure is added.

3.5 Main predictions

Organizational sorting. Conditional on I and E dominating the outside options, low internal manufacturing capability selects entrusted production and high capability selects internal production. Lower entrusted-route friction expands the entrusted region; a higher fixed CMO price contracts it.

MAH-relevant projects and advancement. The reform has a zero effect whenever the newly feasible E value does not exceed the best old route. Otherwise it raises expected project value and, at fixed service price, raises common project advancement. The response is larger where the entrusted value gain is larger; MAH does not change a_i, project quality, or the advancement-cost function.

CMO equilibrium and scarcity. The qualified-capacity price is unique. Reform-induced entrusted demand can raise this price, weakly reducing the direct fixed-price gains in project value and advancement. Fixed-price comparisons and equilibrium-price comparisons are therefore distinct.

Planning, organization, and observed outcomes. At equilibrium,

$$\Lambda_i^{\text{plan}} = a_i x_i^*,$$

while expected realized retained outcomes are

$$Y_i^{\text{ret}} = a_i x_i^* \int s(q)\,[\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}]\,dF(q,m). \tag{17}$$

Thus observed outcomes can change through advancement or route composition. Holder–producer separation is observed only after route assignment and cannot cause the earlier x_i choice.

Novelty composition. For $g \in \{O, \text{Inc}\}$, outcomes use the same x_i^* and the mixture $\rho_g F_g$. The baseline permits either class to have a zero or larger response and imposes no original-versus-incremental ranking. Patent applications are not a baseline endogenous outcome.

4 Extension: Research–Development Allocation and Patenting

The baseline establishes how MAH changes the anticipated value of advancing a viable project, but deliberately does not divide innovation resources between upstream research and downstream development. This section adds that division without changing the route model. The route block continues to supply $\Omega_i(M, p_m)$, and M continues to act directly only through the institutional wedge $\tau_E(M)$. The extension replaces the baseline control x_i with two extension-only controls: upstream research x_i^R and development / project advancement x_i^D.

Let $\bar{X}_i$ denote a fixed innovation-resource ceiling and let $\gamma_i \geq 0$ measure research–development complementarity. Conditional on the anticipated route value Ω_i, the developer solves

$$\begin{aligned} \max_{x_i^R, x_i^D \geq 0} \quad & u_i x_i^R - \frac{\kappa_{Ri}}{2}(x_i^R)^2 + \beta\Omega_i(a_i + \gamma_i x_i^R)x_i^D - \frac{\kappa_{Di}}{2}(x_i^D)^2, \\ \text{subject to} \quad & x_i^R + x_i^D \leq \bar{X}_i. \end{aligned} \tag{18}$$

The planning-stage project mass is $N_i = (a_i + \gamma_i x_i^R) x_i^D$. The resource ceiling represents financing or managerial resources available within the cohort; a firm is constrained only when its KKT multiplier is positive. The appendix gives the full KKT system and assumes the extension-specific curvature condition $\kappa_{Ri}\kappa_{Di} > (\beta\Omega_i\gamma_i)^2$, which makes the problem strictly concave.

The resource regime determines the research response. At an interior slack solution,

$$\frac{\partial x_i^{D,u}}{\partial \Omega_i} > 0, \qquad \frac{\partial x_i^{R,u}}{\partial \Omega_i} \geq 0, \tag{19}$$

where the second inequality is strict when $\gamma_i > 0$ and becomes zero when $\gamma_i = 0$. An unconstrained firm can therefore expand development without cutting research.

At an interior binding solution, $x_i^R + x_i^D = \bar{X}_i$, and the KKT system instead gives

$$\frac{\partial x_i^{R,b}}{\partial \Omega_i} = \frac{\beta[\gamma_i(x_i^{D,b} - x_i^{R,b}) - a_i]}{\kappa_{Ri} + \kappa_{Di} + 2\beta\Omega_i\gamma_i}, \qquad \frac{\partial x_i^{D,b}}{\partial \Omega_i} = -\frac{\partial x_i^{R,b}}{\partial \Omega_i}. \tag{20}$$

Hence, if

$$a_i > \gamma_i(x_i^{D,b} - x_i^{R,b}), \tag{21}$$

the increase in commercialization value shifts a fixed resource toward development: x_i^D rises and x_i^R falls. This is a sufficient development-biased value-shift condition, not an assumed patent sign. If it fails, the research response is unsigned by this mechanism.

Patent applications become an explicit extension outcome through

$$P_i^A = h_i(x_i^R), \qquad h_i'(x_i^R) > 0. \tag{22}$$

Thus a constrained firm satisfying (21) has more development and fewer patent applications after a strict equilibrium increase in Ω_i. A slack firm has more development and weakly more patenting; patenting is locally unchanged when $\gamma_i = 0$. A decline concentrated in low-value applications while granted or core patents remain statistically unchanged is compatible with the extension if the high-value patent margin is locally inelastic to the marginal reallocation. That additional composition condition is not an unconditional model prediction.

CMO scarcity remains part of the result. In the extension, study-related CMO demand replaces $a_i x_i^*$ by N_i^*. Optimal N_i^* is nondecreasing in Ω_i, including across a slack/binding switch, so the demand schedule remains weakly decreasing in the candidate CMO price. The same single market-clearing equation therefore determines the service price. The patent signs above apply to the finite binary MAH comparison only when the equilibrium value gain remains strict; scarcity may attenuate it to zero.

This extension explains the central resource-heterogeneity pattern in Gu (2024): development can rise while patent applications fall for resource-constrained firms, whereas development need not crowd out research for firms with slack resources. It does not assume that observed balance-sheet groups reveal the KKT multiplier, does not equate patent applications with research, and does not build in the paper's original-versus-incremental ranking. The full derivation and boundary cases are in the Technical Appendix.

5 Empirical Predictions and Mapping Boundaries

The deterministic model yields three conditional predictions. First, among projects for which internal and entrusted production dominate transfer and abandonment, developers with weaker internal manufacturing capability are more likely to select retained entrusted production. Second, MAH raises project advancement only when the new entrusted value exceeds the best pre-reform option; the response can be zero for many project–developer pairs. Third, reform-induced demand for qualified manufacturing services can raise p_m^*, attenuating fixed-price gains in expected project value and advancement. These predictions concern commercialization organization and project advancement, not patent generation or scientific productivity.

The empirical distinction is between ex ante advancement and downstream observations. Product-level holder–manufacturer separation can proxy route E only when holder identity, manufacturer identity, product identity, and effective date are aligned. Approval or launch records are realized-product outcomes, not direct observations of x_i. Clinical milestones may proxy project advancement only when their status and timing are measured consistently. Patent applications are a separate patenting margin and are not a baseline endogenous outcome. They become endogenous only in the extension through $P_i^A = h_i(x_i^R)$, conditional on a validated link between upstream research allocation and measured applications.

The timing restriction can be summarized as

$$M \longrightarrow \text{anticipated availability and value of } E \longrightarrow \Omega_i \longrightarrow x_i^*, \tag{23}$$

followed by

$$\begin{aligned} x_i^* &\longrightarrow \text{planning-stage projects} \longrightarrow r_i^* \\ &\longrightarrow \text{observed holder–producer separation} \longrightarrow \text{realized products.} \end{aligned} \tag{24}$$

Observed separation cannot be placed before or treated as a cause of the ex ante x_i choice. The original and incremental classes use the same x_i and the mixture $\rho_g F_g$; either class may have a zero or larger response.

Table 1: Model Objects and Measurable Empirical Interfaces

Model object	Candidate interface	Claim boundary
Predetermined capability a_i	Pre-policy development history or other validated capability measure	Does not separately identify a_i, and no policy shift in a_i is allowed.
Advancement x_i and $a_i x_i$	Consistently measured development-stage activity and planning-stage project events	x_i is not pure clinical effort, patent applications, or an approval count.
Route $r_i^* = E$	Product-level holder differs from manufacturer, with time-consistent identifiers	The observed split is downstream evidence on retained organizational assignment, not the policy treatment itself.
Retained outcome Y_i^{ret}	Approval or launch with developer/holder identity	Combines advancement, project mix, deterministic route assignment, and $s(q)$; it does not identify these primitives separately.
CMO scarcity p_m^*	Service price, or consistently measured capacity, utilization, density, waiting-time, or participation proxy	A nonprice proxy does not identify the price or the supply and demand schedules separately.
Novelty class g	Versioned regulatory classification O or Inc	The classifier creates no x_{ig} and imposes no response ranking.
Extension research and patents x_i^R, P_i^A	Upstream research expenditure/activity and audited patent applications or families	The patent measure is not research itself; the sign requires the extension's resource-regime and value-shift conditions.
Extension resource constraint	Predetermined financing or managerial-resource measures	An observed resource proxy does not identify a positive KKT multiplier; regime classification requires additional evidence or estimation.

These entries are interfaces for future validated data. No data set is asserted here to be available, complete, or identifying. Before use, a source must pass provenance, coverage, key-uniqueness, unit, timing, and construct-validity checks. Entry, route-specific realization, and smooth route choice remain undeveloped extensions. Research-versus-development allocation is now a formal but separate extension rather than a baseline calibration object.

6 Conclusion

MAH changes expected project value by making retained entrusted manufacturing available while preserving authorization-holder responsibility. The route is most relevant when a developer can advance valuable projects but lacks project-compatible internal manufacturing capability. Its equilibrium value also depends on qualified-service scarcity: reform-induced entrusted demand can raise p_m^* and attenuate the fixed-price gain. Project advancement therefore rises only for the MAH-relevant set and need not rise for every developer or novelty class. Observed holder–producer separation and realized products occur after advancement and organizational assignment. Patent applications and upstream scientific research remain outside the baseline. The separate allocation extension shows why a strict commercialization-value gain can raise development while reducing research and patenting when a common resource constraint binds, but not when resources are slack. The model makes no welfare claim. This layered partial-equilibrium structure keeps the baseline focused on commercialization organization, manufacturing mismatch, retained authorization, and CMO scarcity while making the patent boundary an explicit conditional prediction rather than an omission.

References

Acemoglu, Daron and Joshua Linn, "Market Size in Innovation: Theory and Evidence from the Pharmaceutical Industry," *Quarterly Journal of Economics*, 2004, *119* (3), 1049–1090.

Arora, Ashish, Andrea Fosfuri, and Alfonso Gambardella, *Markets for Technology: The Economics of Innovation and Corporate Strategy*, Cambridge, MA: MIT Press, 2001.

Barwick, Panle Jia, Hongyuan Xia, and Tianli Xia, "From Free Rider to Innovator: The Rise of China's Drug Development," NBER Working Paper 34977, National Bureau of Economic Research 2026.

Budish, Eric, Benjamin N. Roin, and Heidi Williams, "Do Firms Underinvest in Long-Term Research? Evidence from Cancer Clinical Trials," *American Economic Review*, 2015, *105* (7), 2044–2085.

China Food and Drug Administration, "Notice on Matters Related to the Marketing Authorization Holder System Pilot," https://www.ccfdie.org/zryyxxw/zcfg/fgdt/webinfo/2016/07/1466184737348633.htm 2016. Accessed 2026-05-20.

Dubois, Pierre, Olivier de Mouzon, Fiona Scott Morton, and Paul Seabright, "Market Size and Pharmaceutical Innovation," *RAND Journal of Economics*, 2015, *46* (4), 844–871.

Finkelstein, Amy, "Static and Dynamic Effects of Health Policy: Evidence from the Vaccine Industry," *Quarterly Journal of Economics*, 2004, *119* (2), 527–564.

Gans, Joshua S. and Scott Stern, "The Product Market and the Market for Ideas: Commercialization Strategies for Technology Entrepreneurs," *Research Policy*, 2003, *32* (2), 333–350.

General Office of the State Council of the People's Republic of China, "Pilot Plan for the Marketing Authorization Holder System," https://www.gov.cn/zhengce/content/2016-06/06/content_5079954.htm 2016. Accessed 2026-05-20.

Gu, Shi, "Production Outsourcing and Innovation: Evidence from China's Pharmaceutical Industry," Working Paper SSRN 4770849, Kellogg School of Management, Northwestern University 2024. August 22, 2024 version.

Jia, Ruixue, Xiao Ma, Jianan Yang, and Yiran Zhang, "Improving Regulation for Innovation: Evidence from China's Pharmaceutical Industry," NBER Working Paper 31976, National Bureau of Economic Research 2023.

National Medical Products Administration, "Drug Administration Law of the People's Republic of China," https://english.nmpa.gov.cn/2019-09/26/c_773012.htm 2019. Accessed 2026-05-18.

_ , "Provisions for Drug Registration," https://english.nmpa.gov.cn/2022-06/30/c_785628.htm 2022. Accessed 2026-05-18.

_ , "Provisions for the Supervision and Administration of Drug Manufacturing," https://english.nmpa.gov.cn/2022-06/30/c_785630_2.htm 2022. Accessed 2026-05-18.

_ , "Provisions on Supervising the Implementation of the Drug Marketing Authorization Holder's Primary Responsibility for Drug Quality and Safety," https://www.nmpa.gov.cn/xxgk/fgwj/xzhgfxwj/20221229192548121.html 2022. Released 2022-12-29; effective 2023-03-01; accessed 2026-05-20.

_ , "Announcement on Strengthening Supervision and Administration of Entrusted Production by Drug Marketing Authorization Holders," https://www.nmpa.gov.cn/xxgk/fgwj/xzhgfxwj/20231023160426145.html 2023. NMPA Announcement No. 132 of 2023; accessed 2026-05-20.

Schmookler, Jacob, *Invention and Economic Growth*, Cambridge, MA: Harvard University Press, 1966.

Teece, David J., "Profiting from Technological Innovation: Implications for Integration, Collaboration, Licensing and Public Policy," *Research Policy*, 1986, *15* (6), 285–305.